RESEARCH PROGRAM · VERIFICATION

The Verification Platform: A Level Research Program E01–E18

Research Program Monograph · Version 1.0.0

The independent Julia platform: blind protocols, SHA pins, cross-language replication; blind reading of ζ zeros by two engines and the fair class-B benchmark.

18/18	649+639	×59 000
VERDICTS PASS	BLIND Z ZEROS	CLASS B

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Research Program: The Verification Platform

The verification program is the independent judge of the program: a Julia platform sharing no code with the core or the processor, blind protocols, SHA pins, cross-language replication. Eighteen verdicts PASS; the next wave — QEC memory, digital twin, integration.

1. Program goals

- re-check the core and the processor with an independent implementation;
- perform blind reading of ζ zeros by two engines up to T = 1000;
- run the fair compiler benchmark (class B);
- build the QEC layer and the digital twin of the machine.

2. Methodology

Detector and validator are separated; outputs are SHA256-pinned before validation; thresholds are pre-registered. Cross-language replication: Julia BigFloat ↔ Python mpmath (0.0 disagreement), numpy replicas (0 counter mismatches). Implementation independence is a design principle, not a circumstance.

3. Executed campaigns E01–E18

Table 3.3. Verification campaigns

Block	Campaigns	Key results
infrastructure	E01–E10	controls, Haar, scaling, replays, noise, code, ALU, cost, readout
ζ reading	E11	the blind ζ reader: pins, validation, zero leaks
optics/nonlocality	E12–E14	interferometer; no-signaling; CHSH
classification	E15	topology ablation
engines	E16	the independent Riemann–Siegel: ≤ 3.8e−15
flagship	E17	blind zeros: 649/649 + 639/639; cross-engine 0.0; N(1000) = 649.0
benchmark	E18	class B: ×59,000; the 2q KAK outside the class (10 vs 3)

4. The O-J roadmap

- O-J1 done: blind reading to T = 1000 (E16–E17);
- O-J2 done: class B (E18);
- O-J3: QEC memory — relies on E07 and V21;
- O-J4: the digital twin of the processor;
- O-J5: the integration campaign; target heights T well above 1000.

5. Honest boundaries

E17 does not prove RH above T = 1000, the simplicity of all zeros, or the absence of off-line zeros for $\gamma > 1000$. E18 claims no superiority outside the ISA class. The blind erratum γ_{12} showed that an error can live in the framework rather than in the detector — therefore references are checked too.

6. Key campaigns in detail

6.1. From infrastructure to flagships

E11 — the blind ζ reader

The platform's first full blind protocol: a detector with no access to the reference, SHA256 pinning, validation after fixing. The campaign exercised the blindness mechanics on a live task — reading zeros of a lower height — and showed zero leaks. All subsequent blind protocols (E16, E17) inherit its scheme.

E16 — the independent Riemann–Siegel

An implementation of the Riemann–Siegel formula (the Arias de Reyna version) without external packages: the main sum plus the integral correction. Agreement with the reference ≤ 3.8e−15 against a threshold of 1e−9; the error floor is 1e−17. The meaning: the platform owns a second independent Z(t) engine, independent of the explicit method.

E17 — blind zeros T=1000

The platform flagship: two independent detectors read zeros up to T = 1000 with no access to the reference; 649/649 (EM) and 639/639 (RS), cross-engine 0.0 bit-exact, N(1000) = 649.0 exactly, grid-shift robustness 0. The blind γ_{12} erratum is built in as a negative control: the detector refuted a hand-made framework constant. The verdict honestly states: RH above T = 1000 is not proved.

E18 — class B

The fair compiler benchmark: one ISA, one criterion, five strong baselines. The outcome: ×59,000 in time at equal accuracy; the 2q boundary 10 versus the KAK optimum 3 — outside the class, declared explicitly. The numpy replication: 0 counter mismatches. The E18 methodology becomes the standard for processor stage P5.

E12–E14 — optics and nonlocality

The interferometry and nonlocality block: the interferometer (E12), the absence of signaling (E13), CHSH (E14). These campaigns independently confirm the processor's V24 preview and give an independent estimate of the correlations of distributed clouds.

7. How to reproduce

```
julia modules/verifier_multilang/experiments/run_all.jl all   # E01–E18 (~2.5 min)
julia modules/verifier_multilang/experiments/verdicts.jl      # verdict summary
# pins and results: modules/verifier_multilang/results/
```

8. Pins and integrity

Every blind output is SHA256-pinned before validation: e17_detector_em.csv, e17_detector_rs.csv, e11_detector_output.csv, the reference e17_zetazero_reference.csv. The pin files sit next to the data; the package validator recomputes the hashes on every build. Changing a pin after validation is impossible without a trace in the history.

9. Further reading

- protocols: experiments/experiment_01_15.md;
- master verdicts: results/verdicts_master.json;
- cross-language validation: results/cross_language_validation.md;
- provenance and limitations: PROVENANCE_AND_LIMITATIONS.md.